

Advanced Image Generation Models

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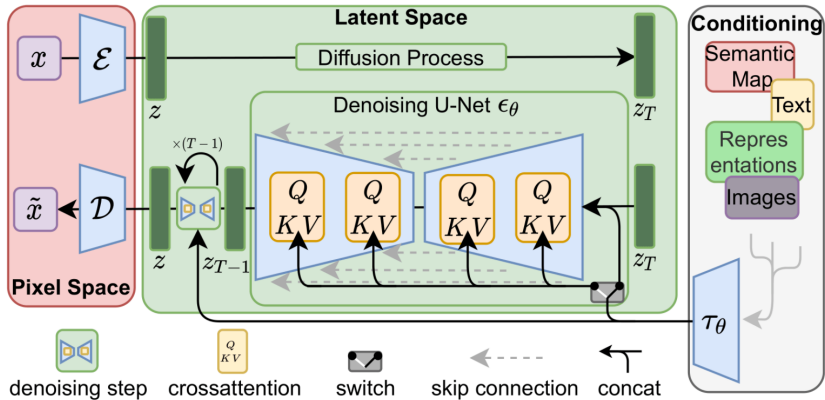


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Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," CVPR 2022, Fig. 3; reproduced in Jay Alammam, "The Illustrated Stable Diffusion", <https://jalammar.github.io/illustrated-stable-diffusion/>

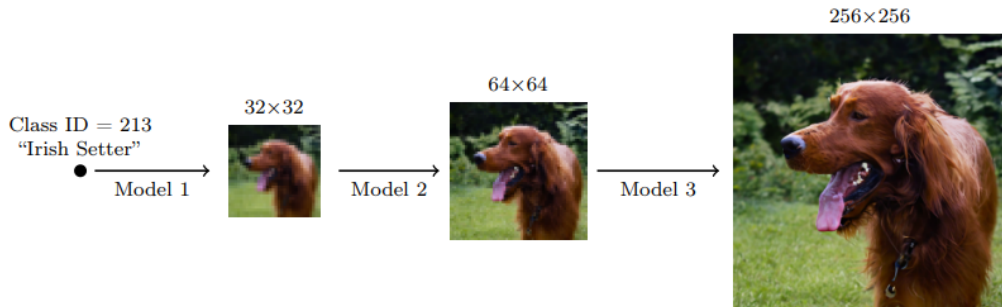
- ▶ Earlier lectures covered why diffusion displaced GANs and how the core denoising recipe works; this lecture pushes beyond the base recipe.

Advanced Image Generation Models

- ▶ *Why it matters:*
 - Diffusion models are reshaping creativity, art, and design.
 - From AI avatars to realistic landscapes, they have endless applications.
 - You'll gain insight into how these models work, and how to use them yourself.

After this session, you will be able to:

- ▶ Describe cascaded diffusion pipelines for high-resolution generation.
- ▶ Summarize score-based diffusion models and their applications (DreamFusion, text-to-SVG, video adapters).
- ▶ Explain how diffusion models can be used to solve PDEs (DiffusionPDE).
- ▶ Utilize text-conditioned diffusion models for controlled image generation.
- ▶ Explain CLIP-guided generation and editing with VQ-GAN and StyleCLIP.



Ho et al., "Cascaded Diffusion Models for High Fidelity Image Generation," *JMLR* 23(47), 2022, Fig. 1.

$$p_{\theta}(x_0) = \int p_{\theta}(z_1)p_{\theta}(z_0 \mid z_1)p_{\theta}(x_0 \mid z_0)dz$$

Ho, Jonathan, Chitwan Saharia, William Chan, David J. Fleet, Mohammad Norouzi, and Tim Salimans. "Cascaded Diffusion Models for High Fidelity Image Generation." *Journal of Machine Learning Research* 23(47):1-33, 2022.

- ▶ Train one base low-resolution model (class-conditional in CDM); the remaining models are super-resolution models conditioned on the previous stage's output.
- ▶ Condition on the low-resolution image, perform bilinear or bicubic upsampling, and concatenate the result to the noised input.
- ▶ Models can be trained independently.
- ▶ Hyperparameters can be tuned specifically for each resolution.

Conditioning Augmentation: Augment the low-res input z

► Method 1: Blurring

- Apply a Gaussian blurring filter
- Effective for models at 128, 256 resolution

► Method 2: Non-truncated Conditioning Augmentation

- Train the super-resolution $p(x|z_0)$ model conditioned on samples from the prior $p(z_0)$ with added corruption from the forward process $q(z_s|z_0)$
- s (corruption / noise strength) is a hyperparameter
- Effective for resolutions < 128

Cascaded Diffusion Models (cont.)



(a) 16×16 base

(b) $16 \times 16 \rightarrow 64 \times 64$ super-resolution, $s = 0$

Ho et al., "Cascaded Diffusion Models for High Fidelity Image Generation," *JMLR* 23(47), 2022, Fig. 9(a,b).

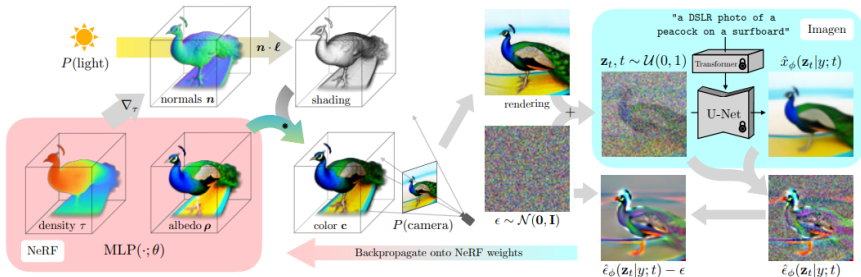
Large improvements in using conditioning augmentation

Conditioning	FID vs train	FID vs validation	IS
No cascading	2.35	2.91	52.72 ± 1.15
16×16→64×64 cascading			
$s = 0$	6.02	5.84	35.59 ± 1.19
$s = 101$	3.41	3.67	44.72 ± 1.12
$s = 1001$	2.13	2.79	54.47 ± 1.05

Ho et al., “Cascaded Diffusion Models for High Fidelity Image Generation,” *JMLR* 23(47), 2022, Table 2(b).

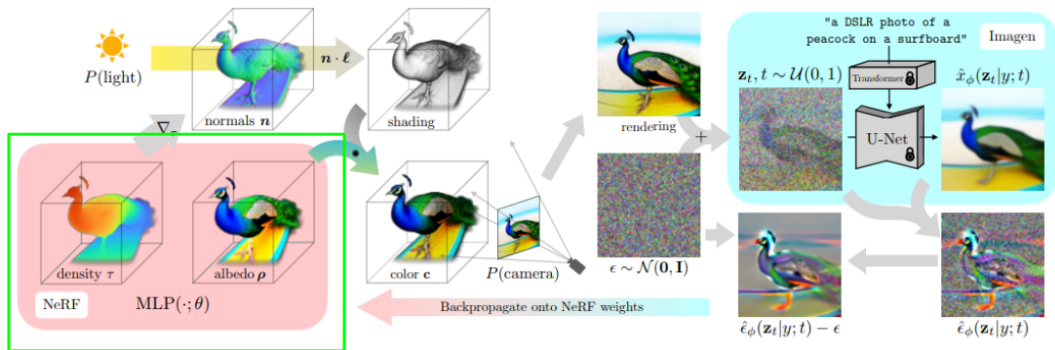
Advanced Diffusion Models: **Score-Based Diffusion Models**

How can you use a pretrained text-image model to generate 3D objects?



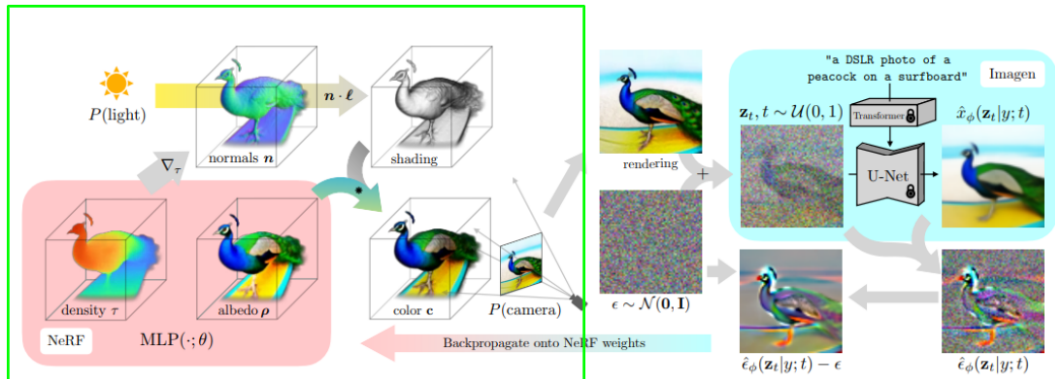
Poole, Ben, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. "DreamFusion: Text-to-3D using 2D Diffusion." *International Conference on Learning Representations (ICLR)*, 2023.

Start with a randomly initialized parameterized 3D representation (NeRF)



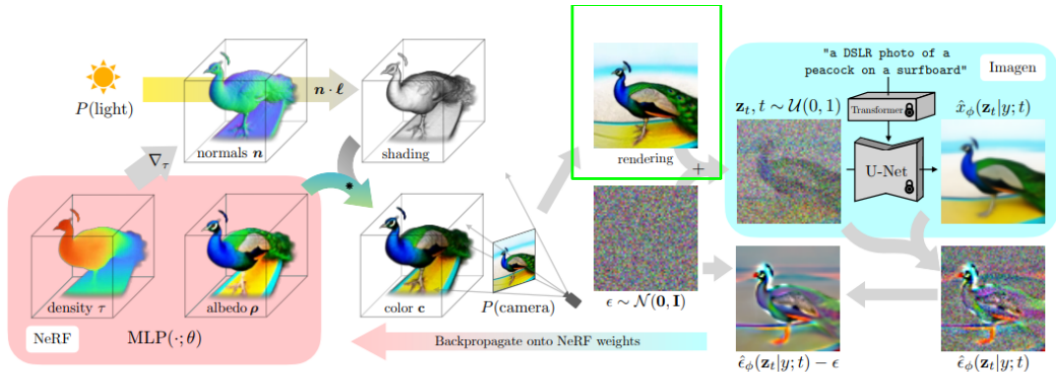
Poole et al., "DreamFusion: Text-to-3D using 2D Diffusion," ICLR 2023, Fig. 3 (annotated).

Randomly sample camera angle and lighting



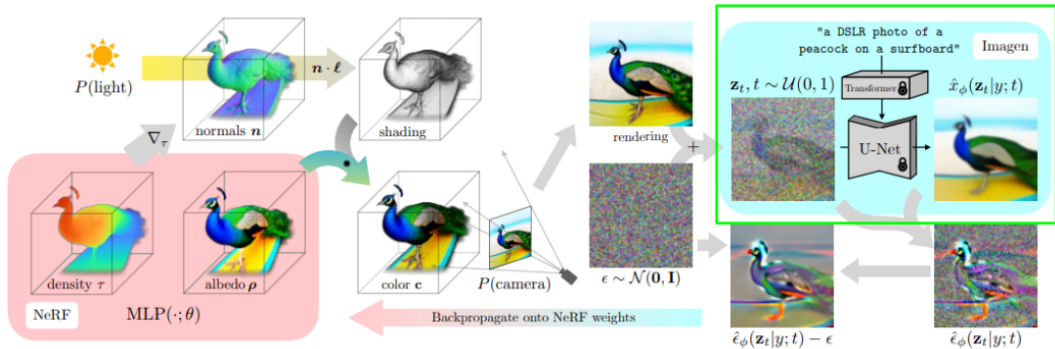
Poole et al., "DreamFusion: Text-to-3D using 2D Diffusion," ICLR 2023, Fig. 3 (annotated).

Render the image (64×64)



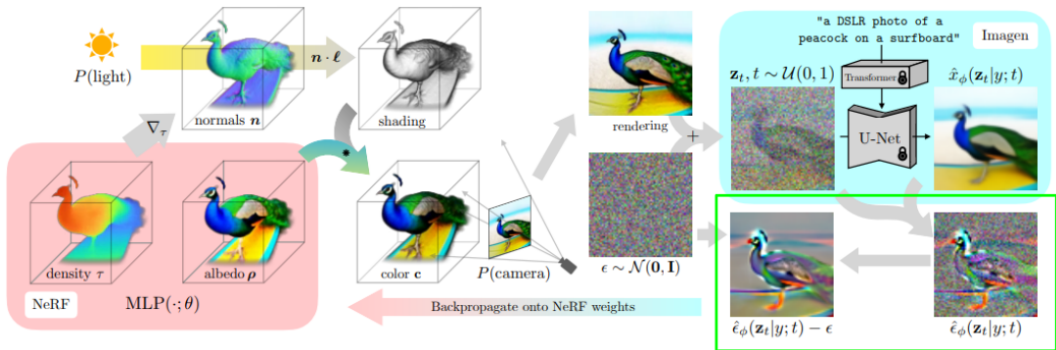
Poole et al., "DreamFusion: Text-to-3D using 2D Diffusion," ICLR 2023, Fig. 3 (annotated).

Noise the image and feed it through the diffusion model to predict the noise



Poole et al., "DreamFusion: Text-to-3D using 2D Diffusion," ICLR 2023, Fig. 3 (annotated).

Compute the Score Distillation Sampling (SDS) loss and backpropagate



Poole et al., "DreamFusion: Text-to-3D using 2D Diffusion," ICLR 2023, Fig. 3 (annotated).

Intuitively, we want the NeRF parameters to produce images that minimize the diffusion loss

$$\mathcal{L}_{\text{Diff}}(\phi, \mathbf{x}) = \mathbb{E}_{t \sim \mathcal{U}(0,1), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [w(t) \|\epsilon_\phi(\alpha_t \mathbf{x} + \sigma_t \epsilon; t) - \epsilon\|_2^2]$$

Poole et al., “DreamFusion,” ICLR 2023, Eq. (1).

$$\nabla_{\theta} \mathcal{L}_{\text{Diff}}(\phi, \mathbf{x} = g(\theta)) = \mathbb{E}_{t, \epsilon} \left[w(t) \underbrace{(\hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t) - \epsilon)}_{\text{Noise Residual}} \underbrace{\frac{\partial \hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t)}{\partial \mathbf{z}_t}}_{\text{U-Net Jacobian}} \underbrace{\frac{\partial \mathbf{x}}{\partial \theta}}_{\text{Generator Jacobian}} \right]$$

Poole et al., “DreamFusion,” ICLR 2023, Eq. (2).

But the objective is generally brittle / expensive (backprop through the U-Net)

It works better to just remove the U-Net Jacobian

$$\nabla_{\theta} \mathcal{L}_{\text{Diff}}(\phi, \mathbf{x} = g(\theta)) = \mathbb{E}_{t, \epsilon} \left[\underbrace{w(t) (\hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t) - \epsilon)}_{\text{Noise Residual}} \underbrace{\frac{\partial \hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t)}{\mathbf{z}_t}}_{\text{U-Net Jacobian}} \underbrace{\frac{\partial \mathbf{x}}{\partial \theta}}_{\text{Generator Jacobian}} \right]$$

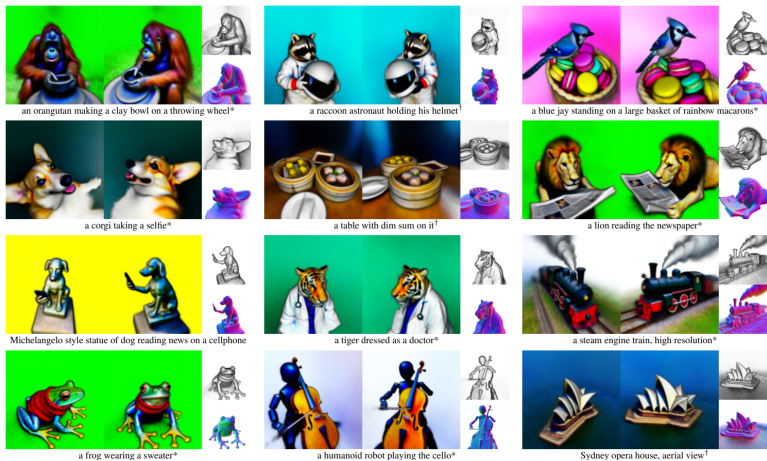
Poole et al., "DreamFusion," ICLR 2023, Eq. (2).

$$\nabla_{\theta} \mathcal{L}_{\text{SDS}}(\phi, \mathbf{x} = g(\theta)) \triangleq \mathbb{E}_{t, \epsilon} \left[w(t) (\hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t) - \epsilon) \frac{\partial \mathbf{x}}{\partial \theta} \right]$$

Poole et al., "DreamFusion," ICLR 2023, Eq. (3).

There is a theoretical motivation (see Appendix in [paper](#))

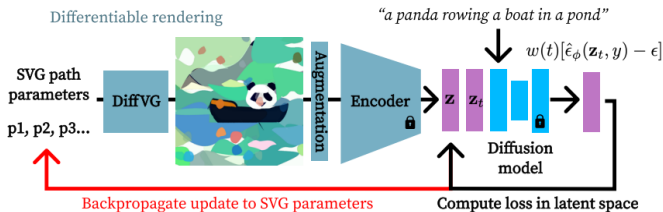
DreamFusion: Text-to-3D Generation (cont.)



Poole et al., "DreamFusion: Text-to-3D using 2D Diffusion," ICLR 2023, Fig. 1.

- ▶ Text-to-SVG generation is a novel application of diffusion models, enabling the creation of vector graphics from textual descriptions.
- ▶ No SVG training data is used: SVG path parameters are optimized through a differentiable rasterizer (DiffVG), guided by a pretrained text-to-image diffusion model.
- ▶ The model generates SVG paths, which are inherently vector-based and scalable without loss of quality.

- ▶ Using SDS loss is not limited to text-to-3D.
- ▶ SDS provides a general way to backpropagate and learn through any differentiable “rendering” pipeline.



Jain, Ajay, Amber Xie, and Pieter Abbeel. “VectorFusion: Text-to-SVG by Abstracting Pixel-Based Diffusion Models.” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1911-1920, 2023.



a train*



an owl standing on a wire*



Underwater Submarine*



a boat*



A photo of a Ming Dynasty vase on a leather topped table.*



A smiling sloth wearing a leather jacket, a cowboy hat and a kilt.*



a tuba with red flowers protruding from its bell*



a blue poison dart frog sitting on a water lily*

Jain, Xie and Abbeel, "VectorFusion: Text-to-SVG by Abstracting Pixel-Based Diffusion Models," CVPR 2023, Fig. 2.

DDPMs as Energy-Based Models (EBMs)

$$p_{\theta}(\tau) \propto e^{-E_{\theta}(\tau)}$$

Yang, Mengjiao, Yilun Du, Bo Dai, Dale Schuurmans, Joshua B. Tenenbaum, and Pieter Abbeel. "Probabilistic Adaptation of Black-Box Text-to-Video Models" (Video Adapter), ICLR 2024, arXiv:2306.01872.

Since the denoising function (ϵ) models the score, we have

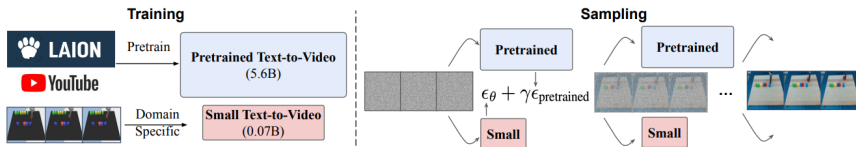
$$\epsilon_{\theta}(\tau^t, t) = \nabla_{\tau} E_{\theta}(\tau^t)$$

Yang et al., "Probabilistic Adaptation of Black-Box Text-to-Video Models" (Video Adapter), ICLR 2024, Sec. 2.

Learn a distribution as a product of a pretrained (large) model and domain specific (small) model

$$p_{\text{product}}(\tau|\text{text}) \propto \underbrace{p_{\text{pretrained}}(\tau|\text{text})}_{\text{Pretrained Prior}} \underbrace{p_{\theta}(\tau|\text{text})}_{\text{Video Model}}.$$

Yang et al., “Probabilistic Adaptation of Black-Box Text-to-Video Models” (Video Adapter), ICLR 2024, Eq. (5).



Yang et al., “Probabilistic Adaptation of Black-Box Text-to-Video Models” (Video Adapter), ICLR 2024, Fig. 2.

Under the product assumption, we can use the EBM interpretation to combine the scores of the diffusion models

$$p_{product}(\tau|\text{text}) \propto e^{-E_1(\tau)} e^{-E_2(\tau)} = e^{-(E_1(\tau)+E_2(\tau))}$$

After Yang et al., “Probabilistic Adaptation of Black-Box Text-to-Video Models” (Video Adapter), ICLR 2024, Sec. 2.

$$\nabla_{\tau} E_{\theta}(\tau) = -(\nabla E_1(\tau) + \nabla E_2(\tau))$$

After Yang et al., “Probabilistic Adaptation of Black-Box Text-to-Video Models” (Video Adapter), ICLR 2024, Sec. 2.

Algorithm 1 Sampling algorithm of Video Adapter

Input: Pretrained Text-to-Video Model $\epsilon_{\text{pretrained}}(\tau, t|\text{text})$, Inverse Temperature α , Prior strength γ .

Initialize sample $\tau_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

for $t = T, \dots, 1$ **do**

$\tilde{\epsilon}_{\text{text}} \leftarrow \epsilon_{\theta}(\tau_t, t|\text{text}) + \gamma \epsilon_{\text{pretrained}}(\tau_t, t|\text{text})$

 // compute score using text-conditioned prior

$\epsilon \leftarrow \epsilon_{\theta}(\tau_t, t)$

 // compute unconditional score

$\tilde{\epsilon}_{\text{cfg}} \leftarrow \epsilon + \alpha(\tilde{\epsilon}_{\text{text}} - \epsilon)$

 // compute classifier guidance score

$\tau_{t-1} = \text{ddpm_sample}(\tau_t, \tilde{\epsilon}_{\text{cfg}})$

 // run diffusion sampling (can use other samplers)

end for

Yang et al., “Probabilistic Adaptation of Black-Box Text-to-Video Models” (Video Adapter), ICLR 2024, Algorithm 1.

Model	Bridge			Ego4D		
	FVD ↓	FID ↓	Param (B) ↓	FVD ↓	IS ↑	Param (B) ↓
Small (S)	186.8	38.8	0.07	228.3	2.28	0.07
Small (S) + Pretrained	177.4	37.6	0.07	156.3	2.82	0.07
Small (L)	152.5	30.1	0.14	65.1	3.31	2.8
Small (L) + Pretrained	148.1	29.5	0.14	52.5	3.53	2.8
Pretrained	350.1	42.6	5.6	91.7	3.12	5.6

Yang et al., “Probabilistic Adaptation of Black-Box Text-to-Video Models” (Video Adapter), ICLR 2024, Table 1.

Can adapt to different domains (robot trajectories, ego-centric camera)



Video Adapter sample, Bridge robot data (Yang et al., ICLR 2024). [Link: GIF](#)



Video Adapter sample, Ego4D egocentric data (Yang et al., ICLR 2024). [Link: GIF](#)

Advanced Diffusion Models: **PDE-based Diffusion Models**

- ▶ Use diffusion models, originally built for image generation, to solve and analyze PDEs.
- ▶ Learn complex PDE relationships by adding noise to solutions and reversing the process.
- ▶ Effective for handling incomplete or uncertain data.
- ▶ Valuable in scientific and engineering applications.

Core Idea

- ▶ Diffusion models transform data into a simple distribution (e.g., Gaussian noise) via a forward process.
- ▶ A reverse process is learned to reconstruct the original data from noise.
- ▶ For PDEs, the model maps input or boundary conditions to PDE solutions, capturing the equation's dynamics.

- ▶ **Generative Solutions:** Can produce multiple plausible PDE solutions, useful under uncertainty or incomplete information.
- ▶ **Uncertainty Modeling:** Capture a distribution of possible solutions, not just a single answer.
- ▶ **Partial Data Learning:** Train effectively on datasets with missing or incomplete data, suitable for real-world scenarios.
- ▶ **Flexibility:** Adaptable to both forward (solving PDEs) and inverse (parameter estimation) problems.

The diffusion equation is a fundamental PDE describing the distribution of a quantity (such as heat or particles) diffusing through a medium:

$$\frac{\partial u}{\partial t} = D \nabla^2 u \quad (1)$$

where:

- ▶ $u = u(\mathbf{x}, t)$ is the quantity of interest (e.g., concentration, temperature)
- ▶ D is the diffusion coefficient
- ▶ ∇^2 is the Laplacian operator

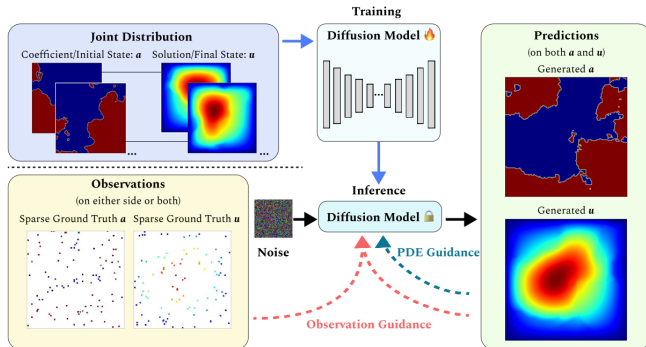


Figure 1: We propose DiffusionPDE, a generative PDE solver under partial observations. Given a family of PDE with coefficient (initial state) a and solution (final state) u , we train the diffusion model on the joint distribution of a and u . During inference, we gradually denoise a Gaussian noise, guided by sparse observation and known PDE function, to recover the full prediction of both a and u that align well with the sparse observations and the given equation.

Huang et al., "DiffusionPDE: Generative PDE-Solving Under Partial Observation," NeurIPS 2024, Fig. 1.

Algorithm 1 Sparse Observation and PDE Guided Diffusion Sampling Algorithm.

```

1: input DeterministicSampler  $D_\theta(\mathbf{x}; \sigma)$ ,  $\sigma(t_{i \in \{0, \dots, N\}})$ , TotalPointCount  $m$ , ObservedPointCount  $n$ , Observation  $\mathbf{y}$ , PDEFunction  $f$ , Weights  $\zeta_{obs}, \zeta_{pde}$ 
2: sample  $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \sigma(t_0)^2 \mathbf{I})$  ▷ Generate initial sampling noise
3: for  $i \in \{0, \dots, N-1\}$  do
4:    $\hat{\mathbf{x}}_N^i \leftarrow D_\theta(\mathbf{x}_i; \sigma(t_i))$  ▷ Estimate the denoised data at step  $t_i$ 
5:    $\mathbf{d}_i \leftarrow (\mathbf{x}_i - \hat{\mathbf{x}}_N^i) / \sigma(t_i)$  ▷ Evaluate  $d\mathbf{x}/d\sigma(t)$  at step  $t_i$ 
6:    $\mathbf{x}_{i+1} \leftarrow \mathbf{x}_i + (\sigma(t_{i+1}) - \sigma(t_i))\mathbf{d}_i$  ▷ Take an Euler step from  $\sigma(t_i)$  to  $\sigma(t_{i+1})$ 
7:   if  $\sigma(t_{i+1}) \neq 0$  then
8:      $\hat{\mathbf{x}}_N^i \leftarrow D_\theta(\mathbf{x}_{i+1}; \sigma(t_{i+1}))$  ▷ Apply 2nd order correlation unless  $\sigma = 0$ 
9:      $\mathbf{d}'_i \leftarrow (\mathbf{x}_{i+1} - \hat{\mathbf{x}}_N^i) / \sigma(t_{i+1})$  ▷ Evaluate  $d\mathbf{x}/d\sigma(t)$  at step  $t_{i+1}$ 
10:     $\mathbf{x}_{i+1} \leftarrow \mathbf{x}_i + (\sigma(t_{i+1}) - \sigma(t_i)) (\frac{1}{2}\mathbf{d}_i + \frac{1}{2}\mathbf{d}'_i)$  ▷ Apply the trapezoidal rule at step  $t_{i+1}$ 
11:   end if
12:    $\mathcal{L}_{obs} \leftarrow \frac{1}{n} \|\mathbf{y} - \hat{\mathbf{x}}_N^i\|_2^2$  ▷ Evaluate the observation loss of  $\hat{\mathbf{x}}_N^i$ 
13:    $\mathcal{L}_{pde} \leftarrow \frac{1}{m} \|\mathbf{0} - f(\hat{\mathbf{x}}_N^i)\|_2^2$  ▷ Evaluate the PDE loss of  $\hat{\mathbf{x}}_N^i$ 
14:    $\mathbf{x}_{i+1} \leftarrow \mathbf{x}_{i+1} - \zeta_{obs} \nabla_{\mathbf{x}_i} \mathcal{L}_{obs} - \zeta_{pde} \nabla_{\mathbf{x}_i} \mathcal{L}_{pde}$  ▷ Guide the sampling with  $\mathcal{L}_{obs}$  and  $\mathcal{L}_{pde}$ 
15: end for
16: return  $\mathbf{x}_N$  ▷ Return the denoised data

```

Huang et al., “DiffusionPDE: Generative PDE-Solving Under Partial Observation,” NeurIPS 2024, Algorithm 1.

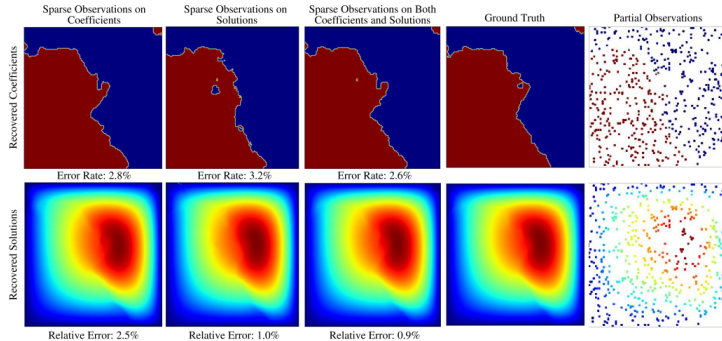


Figure 2: Different from forward and inverse PDE solvers, DiffusionPDE can take sparse observations on either the coefficient a or the solution u to recover both of them, using one trained network. Here, we show the recovered a and u of the Darcy's equation given sparse observations on a , u , or both. Compared with the ground truth, we see that our method successfully recovers the PDE in all cases.

Huang et al., "DiffusionPDE: Generative PDE-Solving Under Partial Observation," NeurIPS 2024, Fig. 2.

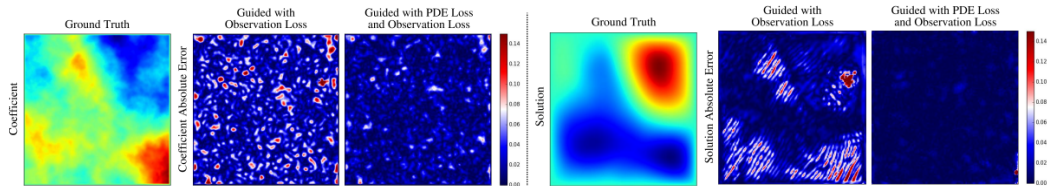


Figure 3: Usefulness of PDE loss. We visualize the absolute errors of the recovered coefficient and solution of the Helmholtz equation with and w/o PDE loss. We compare having only the observation loss with applying the additional PDE loss. The errors drop significantly when using PDE loss.

Huang, Jiahe, Guandao Yang, Zichen Wang, and Jeong Joon Park. "DiffusionPDE: Generative PDE-Solving Under Partial Observation." *Advances in Neural Information Processing Systems* (NeurIPS), 2024.

Advanced Diffusion Models: **Text-Conditioned**

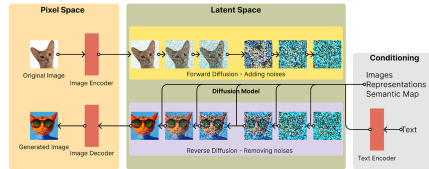


Diagram after Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," CVPR 2022, Fig. 3.

- ▶ **Overview:** Text-conditioned diffusion models enable the generation of highly specific images from textual prompts, such as *“a cat wearing sunglasses standing in Times Square”*. These models use a pretrained text encoder, such as CLIP or a transformer language model, to extract semantic representations from the input text.
- ▶ **Text Conditioning:**
 - The text encoder maps the prompt y to an embedding $e_y = \text{TextEncoder}(y)$.
 - This embedding is injected into the denoising U-Net via **cross-attention**: the general attention layer covered earlier, with the queries taken from the image feature map and the keys and values taken from e_y . Each spatial position therefore attends to the tokens of the prompt.
 - The details of where these layers sit inside the U-Net are covered later.
- ▶ **The only change to the diffusion recipe:** everything from the base denoising recipe covered earlier carries over; the noise predictor simply takes the text embedding as an extra input, $\epsilon_\theta(x_t, t, e_y)$, in both the training objective and every denoising step.

- **Effect of Conditioning:** By conditioning the denoising process on e_y , the model generates images that faithfully reflect the textual description. Cross-attention ensures that relevant parts of the text influence corresponding regions in the image, enabling complex and coherent scene synthesis.

Advanced Image Generation:

CLIP-Guided Generation and Editing

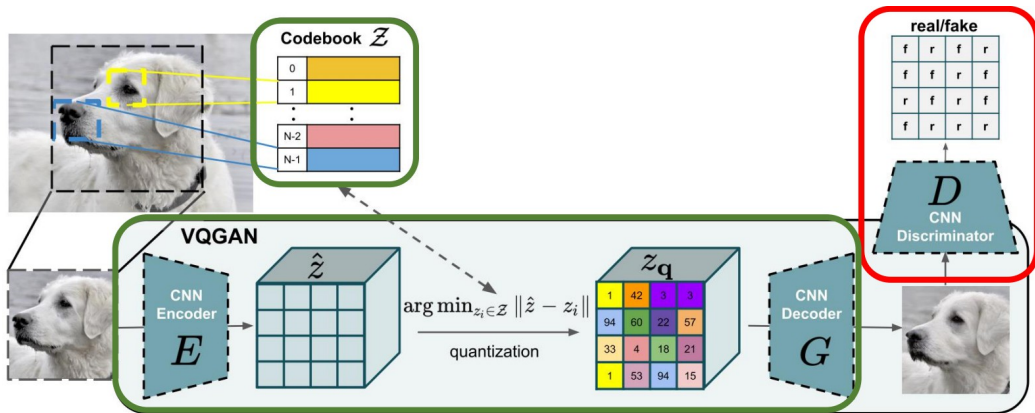
VQ-GAN: Vector Quantized Generative Adversarial Network

VQ-GAN is a generative model that combines the strengths of vector quantization and GANs to generate high-quality images.

- ▶ **Vector Quantization:** Encodes images into discrete latent codes, allowing for efficient representation and generation.
- ▶ **GAN Framework:** Uses a discriminator to guide the generator in producing realistic images.
- ▶ **Applications:** Image synthesis, super-resolution, and style transfer.

Architecture: VQ-GAN consists of an encoder, a quantizer, a decoder, and a discriminator.

- ▶ **Encoder:** Maps input images to a continuous latent space.
- ▶ **Quantizer:** Discretizes the continuous latent space into a finite set of codes.
- ▶ **Decoder:** Reconstructs images from the quantized latent codes.
- ▶ **Discriminator:** Evaluates the realism of generated images, providing feedback to the generator.



VQ-VAE + GAN

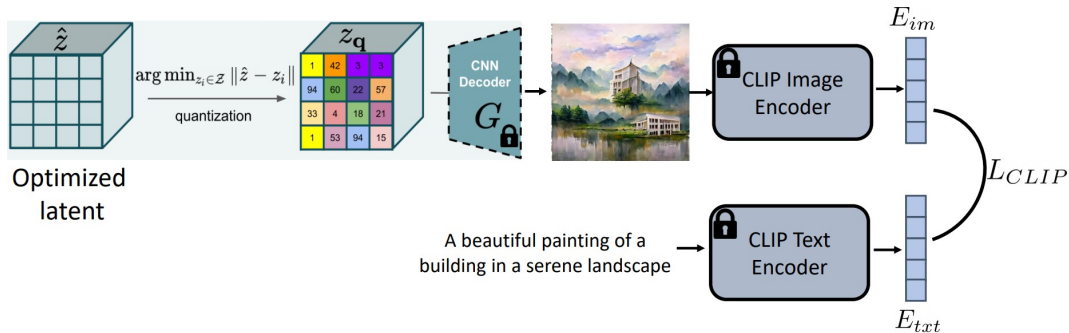
[Taming Transformers for High-Resolution Image Synthesis, Esser, Rombach & Ommer, CVPR 2021]

VQ-GAN + CLIP pairs two models so that vector quantization supplies the image generator and contrastive language-image pretraining supplies the text-based objective.

- ▶ **VQ-GAN:** A generative model that uses vector quantization to encode images into discrete latent codes, enabling efficient representation and high-quality image synthesis.
- ▶ **CLIP:** A model that learns visual representations by aligning images and text, allowing for zero-shot classification and improved understanding of visual content.

How They Work Together:

- ▶ **Image Generation:** VQ-GAN generates images by decoding quantized latent codes, while CLIP provides a text-based conditioning mechanism that guides the generation process.
- ▶ **Text Conditioning:** CLIP's text embeddings can be used to condition the VQ-GAN generator, allowing it to produce images that match specific textual descriptions.
- ▶ **Loss Function:** The similarity between generated image embeddings and text embeddings from CLIP can be used as a loss function to optimize the VQ-GAN generator, ensuring that the generated images align with the given text prompts.



$$L_{CLIP} = 1 - \text{Cos}(E_{im}, E_{txt})$$

[VQGAN-CLIP: Open Domain Image Generation and Editing with Natural Language Guidance, Crowson et al., ECCV 2022]

Applications:

- ▶ **Text-to-Image Synthesis:** VQ-GAN + CLIP can generate images based on textual descriptions, enabling creative applications like art generation and design.
- ▶ **Image Editing:** The combination allows for editing images based on text prompts, such as changing the style or content of an image while preserving its structure.
- ▶ **Zero-Shot Learning:** CLIP's zero-shot capabilities classify generated images into categories without additional training.



"A sketch of 3D printer by da Vinci"



"An autogyro flying, artstation"



"A futuristic city in synthwave style"



"A colored pencil drawing of a waterfall"



"A painting of a city in a deep valley"

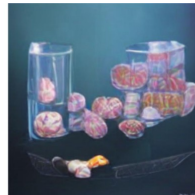


"Baba Yaga's house, fantasy art"

[VQGAN-CLIP, Crowson et al., ECCV 2022, Fig. 2]

Generation

A beautiful
painting of a
building in a
serene landscape



Editing



"A cake made of ice"



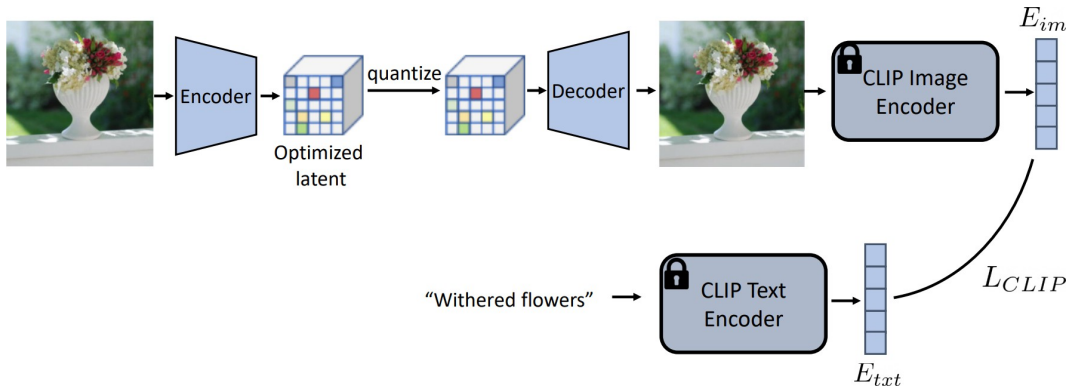
[Composite slide figure; generated image from VQGAN-CLIP, Crowson et al., ECCV 2022, Fig. 2]

Editing needs no new machinery: only the starting point changes.

- ▶ **Initialization is the whole difference:** for generation the initial image is random pixel values; for editing it is the image we want to modify. The architecture and the loss are unchanged.
- ▶ **The prompt is used identically:** the text describing the desired change enters the CLIP similarity loss exactly as a generation prompt would.
- ▶ **Structure is retained** because the optimization starts from the source image. Where a region must be held fixed, the gradients on the corresponding parts of the latent vector can be zeroed out (masking).
- ▶ **Applications:** recolouring, weather and texture changes, and style transfer on real photographs, not only synthesis from scratch.

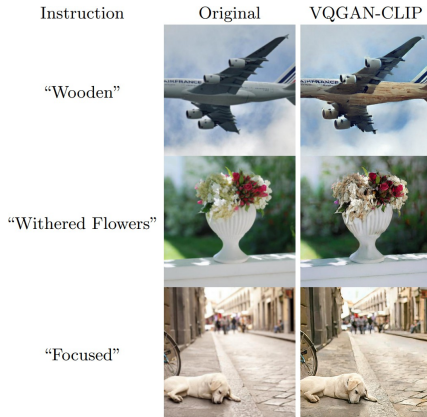
[VQGAN-CLIP, Crowson et al., ECCV 2022, Sections 2.5 and 4]

VQ-GAN + CLIP Editing (cont.)



[VQGAN-CLIP: Open Domain Image Generation and Editing with Natural Language Guidance, Crowson et al., ECCV 2022]

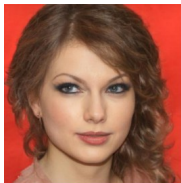
VQGAN + CLIP Editing (cont.)



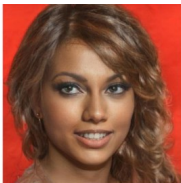
[VQGAN-CLIP: Open Domain Image Generation and Editing with Natural Language Guidance, Crowson et al., ECCV 2022, Fig. 7]

StyleCLIP combines CLIP (Contrastive Language-Image Pretraining) with StyleGAN to generate images based on text prompts, allowing for fine-grained control over image attributes.

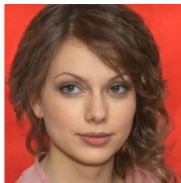
- ▶ **CLIP Embeddings:** Uses CLIP to encode text prompts into embeddings that capture semantic information.
- ▶ **StyleGAN Manipulation:** Applies these embeddings to manipulate the latent space of StyleGAN, enabling the generation of images that match the desired attributes described in the text.
- ▶ **Applications:** Useful for tasks like image editing, attribute manipulation, and creative content generation.
- ▶ **Benefits:** Generates diverse images from textual descriptions without retraining StyleGAN.
- ▶ **Costs:** Bound to the attributes StyleGAN already models; edits outside its latent space either fail or introduce artifacts.



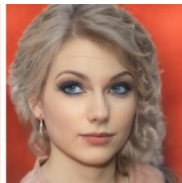
Input



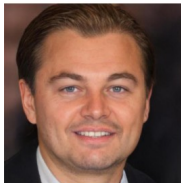
"Beyonce"



"A woman without
makeup"



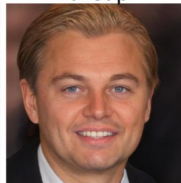
"Elsa from Frozen"



Input



"A man with
..."



"A blonde man"



"Donald Trump"

[StyleCLIP: Text-Driven Manipulation of StyleGAN Imagery, Patashnik et al., ICCV 2021]



[StyleCLIP, Patashnik et al., ICCV 2021, Fig. 8 (AFHQ dogs)]

Input



Trees



Clouds



Spires



Round Roof



[StyleCLIP, Patashnik et al., ICCV 2021, Fig. 19 (LSUN Church)]

Advanced Diffusion Models: **Limitations**

Challenge	Explanation
Compute cost	Denoising many steps can be slow / costly
Biases & ethics	Models reflect biased training data
Guidance artifacts	High guidance can oversharpen or distort
Fine-tune data needs	Requires quality, diverse data
Legal concerns	Generated content may face IP issues

Advanced Diffusion Models: **References**

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